

Project Overview

- **Project Name:** Sustain Eats: "Making Every Bite Count"
- **Project Domain:** Food
- **Project Team:** Apurva Pingale, Chirag Patil, Shreenidhi Pai, Shweta Rane
- **Start Date:** 08/20/2024
- **End Date:** 11/14/2024
- **Project Goal:**

The goal of **Sustain Eats** is to create a software solution that helps event organizers, administrators, and participants track food consumption and minimize food waste at events. By monitoring attendee participation, food preferences, and leftover quantities, **Sustain Eats** aims to provide event organizers, and administrators with actionable insights that allow them to make more informed food procurement decisions for future events. These insights will enable event planners to make data-driven decisions about future food planning, ensuring that food is purchased more accurately to meet actual needs and preferences, while minimizing waste.

Task List:

Phase 1: Problem Statement

Task No.	Task title	Description	Assigned to	Completed in (hrs)
1.	Selecting the business domain and general problem description	Select the domain (Travel, Food, Entertainment, Retail) and research and define the scope of the business problem. Describe the business from the owner's perspective.	All	3
2.	Data to Be Captured	Identify the key data elements required for the business. Define the main participants and how they interact.	All	10
3.	Identify Key Participants	List and define the major participants involved in the business including details of their roles and interactions.	All	10
4.	Business Goals and Functional Requirements	Develop business goals or functional requirements for the	Chirag, Shweta	15

		system, focusing on reports and aggregate information needed to analyze and grow the business.		
5.	Report Requirements and Insights	Define the types of reports the business owner would like to see.	Apurva, Shreenidhi	3
6.	Review Requirements Data	Review the identified data to make sure it covers all business needs.	All	5
7.	Submission	Summarize everything into a clear and organized document for submission.	Chirag	3

Phase 1: Problem Statement (Revised)

Task No.	Task title	Description	Assigned to	Completed in (hrs)
1.	Review Initial Problem Statement	Review the original problem statement to identify areas of improvement.	All	1
2.	Analyze Feedback for Revisions	Analyze the feedback and identify the changes to be made in the problem statement.	Chirag, Apurva	2
3.	Revise Problem Statement	Modify the original problem statement based on the feedback.	Shreenidhi, Shweta	4
4.	Review Revised Problem Statement and submit	Review the revised problem statement to ensure all necessary changes have been made and that it accurately captures the business problem.	All	3
5.	Submission	Prepare the zip folder based on the guidelines given.	Chirag	1

Phase 2: EER Modeling

Task No.	Task title	Description	Assigned to	Completed in (hrs)
1.	Review Revised Problem Statement	Review the final revised problem statement from Phase 1 before starting the EER diagram.	All	1
2.	Identify Key Entities	Identify and define at least 5 entities from the problem domain	Shreenidhi, Apurva	1
3.	Identify Key Relationships	Identify and define at least 5 relationships between entities, including a relationship that spans more than 2 entities.	Chirag, Shweta	5
4.	Define Cardinalities (Min, Max)	Define the cardinality for each relationship using (min, max)	Shweta, Shreenidhi	7
5.	Define Keys for Entities	Identify the primary key for each entity	Chirag, Apurva	3
6.	Identify Composite/Multi-valued/Composite multi-valued attributes	Identify the composite/multi-valued/composite multi-valued attributes	Chirag, Apurva	3
7.	Identify Weak Entities and Relationships	Identify if any weak entities and weak relationships are required	Shweta, Shreenidhi	2
8.	Create EER Diagram	Draw the EER diagram by hand making sure all entities, relationships, cardinalities, keys, and attributes are represented.	Shreenidhi, Apurva	7
9.	Review and Finalize the EER Diagram	Review the EER diagram ensuring all required entities, relationships, and attributes are included. Validate that the diagram follows the guidelines.	All	5
10.	Submission	Prepare the zip folder based on the guidelines given.	Apurva	1

Phase 3: Mapping to Relations

Task No.	Task title	Description	Assigned to	Completed in (hrs)
1.	Review the EER Diagram from Phase 2	Revisit the final EER diagram from Phase 2 before starting the mapping process.	Chirag, Shweta	1
2.	Convert Entities into Relations	Convert each entity from the EER diagram into a relation. Identify attributes and assign them to the correct relations.	Shreenidhi, Apurva	7
3.	Convert Relationships into Relations	Convert the relationships from the EER diagram into relational format. Identify foreign keys and create new tables if needed	Shweta, Chirag	7
4.	Identify Primary Keys for Relations	Identify the primary key for each relation.	Shreenidhi, Apurva	1
5.	Identify Foreign Keys	Identify and list the foreign keys for each relation based on the relationships in the EER diagram.	Apurva, Shweta	2
6.	Identify Candidate Keys	For each relation, identify all candidate keys.	Chirag, Shreenidhi	2
7.	Identify Functional Dependencies	Identify the functional dependencies for each relation.	Chirag, Shweta	5
8.	Review and Validate the Mapped Relations	Review all the relations and validate all primary keys, foreign keys, candidate keys, and functional dependencies.	All	7
9.	Submission	Prepare the subfolders for all three phases and zip folder based on the guidelines given.	Shreenidhi	2

Phase 4: DB Creation, Complex SQL Queries (Demo Phase)

Task No.	Task title	Description	Assigned to	Completed in (hrs)
1.	Normalize all relations up to BCNF	Review all relations after they have been normalized to BCNF. Ensure all functional dependencies are captured.	Chirag, Shweta	8
2.	Write DDL Script for Table Creation (projectDBcreate.sql)	Write the DDL script (projectDBcreate.sql) to create all the tables that have to be named as per the given format. This includes the tables with their constraints, (and views, indexes, and other objects if you have them.). Ensure tables are created in the correct order.	Shreenidhi, Apurva	7
3.	Write DML Script for Data Insertion (projectDBinsert.sql)	Write the DML insert script (projectDBinsert.sql) to insert about 40-50 rows of data into each table. Ensure the data inserted makes sense with respect to the primary and foreign keys.	Shweta, Shreenidhi	10
4.	Write DML Script for Updates (projectDBupdate.sql)	Write the DML update script (projectDBupdate.sql) to perform a series of inserts, deletes, and updates on the tables. This will allow testing of how the database responds to changes. Ensure the updates affect the results of the queries.	Chirag, Apurva	6
5.	Write Drop Script (projectDBdrop.sql)	Write the DROP script (projectDBdrop.sql) that will clean up the database by dropping all created tables. Ensure tables are dropped in the reverse order of creation to maintain referential integrity.	Chirag, Shreenidhi	3

6.	Write Ad-Hoc SQL Queries (projectDBqueries.sql)	Write at least 7 ad-hoc SQL queries (projectDBqueries.sql). Use the comment facility in sqlplus (starting a line with --) to write the English version (IMPORTANT) of your query, followed by the SQL version of the query	Shweta, Apurva	4
7.	Review SQL Scripts	Review all SQL scripts to ensure there are no errors. Make sure they execute correctly in Oracle. Run the DDL, DML, and drop scripts in sequence to verify correctness.	Shreenidhi, Apurva	8
8.	Test Ad-Hoc SQL Queries on Populated Database	Run the ad-hoc SQL queries on the populated database and ensure that the output is as expected.	Chirag, Shweta	5
9.	Prepare Final Team Task Management Document	Prepare the final task management document that outlines how tasks were allocated and managed during the project, highlighting contributions from each team member for all four phases.	Shweta	3
10.	Submission	Prepare the subfolders for all four phases and zip folder based on the guidelines given.	Chirag	2